

## Department of Mathematical Sciences

Rajiv Gandhi Institute Of Petroleum Technology, Jais

End Semester Examination (AY 2023 - 24 / Even / Sem - II)

B.Tech. in M&amp;C

Engineering Practices in Mathematics and Computing (MA 122)

Duration : 3 hours

May 2, 2024

Total Marks : 50

**Instructions.**

- |                                           |                                |                                         |
|-------------------------------------------|--------------------------------|-----------------------------------------|
| i) Attempt all the questions using MATLAB | ii) Calculator is NOT allowed. | iii) Symbols have their usual meanings. |
|-------------------------------------------|--------------------------------|-----------------------------------------|

1. State whether the following statements are True or False:

(1 × 5)

- Indentation is required in MATLAB to define code blocks.
- The `eye()` function in MATLAB creates a diagonal matrix with ones on the diagonal.
- An algorithm is a series of logical steps that solves one specific problem.
- The word `true` is a valid logical expression.
- When moving a block of data in the form of specified rows and columns from array A to array B, the shape of the block in A must match the shape of the block in B.

2. Fill in the blanks:

(1 × 5)

- Arithmetic operations can be performed collectively on the individual components of two arrays as long as both arrays \_\_\_\_ or one of them is \_\_\_\_.
- The normal precedence of operators can be overruled by the use of \_\_\_\_.
- A while loop can be repeated a number of times, depending on the \_\_\_\_ being processed.
- MATLAB provides a function called \_\_\_\_ to generate a mesh grid for 3D plotting.
- The function used to find the determinant of a matrix in MATLAB is \_\_\_\_.

3. Answer with proper justifications:

(2 × 5)

- Find the indefinite integral of the function  $f(x) = x^2 + 3x + 2$  using MATLAB's symbolic toolbox.
- Suppose  $X$  is a new variable, with the command,  $X = [-10 : -1 : -15; -2 : 3]$ . What will be the size of  $X$ ?
- What is the difference between `clc` and `clear` commands?
- Determine the solution for the given system of equations using both matrix left division and the inverse matrix method:
 
$$\begin{aligned} 5x + 3y - z &= 10 \\ 3x + 2y + z &= 4 \\ 4x - y + 3z &= 12. \end{aligned}$$
- Create a 3x3 matrix in MATLAB such that each element is the product of its row and column index.

4. Find the mistake in the following commands/codes, if any, and correct them:

(1.5 × 6)

- $X[1,2] = 4$
- $P = \text{ones}(1;3)$

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iii) a = 5
     b = 3
     result = a + b;
     disp('The result is: ', result);
iv) x = 1:5;
     y = x.^2;
     plot(x, y);
v) a = 1;
    b = 2;
    c = a*b;
    print(c);
vi) x=[1 3 5 8 10 12]
     plot(x);

```

5. Write code in MATLAB:

(3.5 × 6)

- i) Write a code to add the given numbers till the number entered is 0, using while loop.
- ii) Write a code using if statement to find the root of a quadratic equation  $ax^2 + bx + c = 0$ , where the inputs  $a$ ,  $b$ ,  $c$  are given by the user. Compute also the CPU time.
- iii) Write a code to find the 100 th prime number.
- iv) Write a code to solve the following differential equation using MATLAB's Symbolic Math Toolbox:

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0,$$

with initial conditions  $y(0) = 1$  and  $y'(0) = 0$ .

- v) Write a code to solve the initial value problem using 'ode45'

$$\frac{d^2y}{dt^2} + y = 0, \quad 0 < t < 5, \quad y(0) = 1, \quad y'(0) = 0.$$

- vi) Write a code to find the minimum value in an array of 100 numbers using for loop.

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